

RecoverAI Joint

Objective recovery intelligence for hip & knee replacement patients — at home.

Two low-cost motion sensors + a daily symptom check-in become a personalised, explainable recovery trajectory — so orthopaedic teams see who needs earlier review.

TEAM ONEHEALTH

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UN SDG 3 · Good Health



UN SDG 9 · Innovation



Submission · 26 Jul 2026

Vision & Mission



VISION

A world where every joint-replacement patient recovers safely at home — and no early warning sign is missed between clinic visits.



MISSION

To turn low-cost wearable movement data and daily symptoms into an explainable recovery trajectory that helps orthopaedic teams review the right patient at the right time.

AIMS & GOALS



Make recovery measurable

Objective knee range-of-motion and function — not questionnaires alone.



Catch off-track recovery early

Multi-signal, explainable alerts before problems escalate.



Reduce clinician burden

A prioritised queue and fewer low-value contacts, not more admin.



Stay affordable & device-agnostic

Reusable sensor pod, phone-based, NHS-ready.

Recovery happens at home — but care can't see it



MEET MARGARET, 68

After a total knee replacement, Margaret goes home in under three days.

For the next six weeks she does her rehab exercises alone. Her physiotherapist sees her only at occasional appointments — and can't tell whether her progress is normal until something has already gone wrong.

90%+ of recovery now happens outside the hospital.



THE PATIENT

Pain, stiffness and mobility are managed between episodic contacts — it's hard to know whether progress is expected.



THE PHYSIOTHERAPIST

Limited visibility between appointments; large caseloads make proactive review difficult.



THE ORTHOPAEDIC SERVICE

Shorter stays push recovery into the community, while teams must avoid unnecessary contacts.

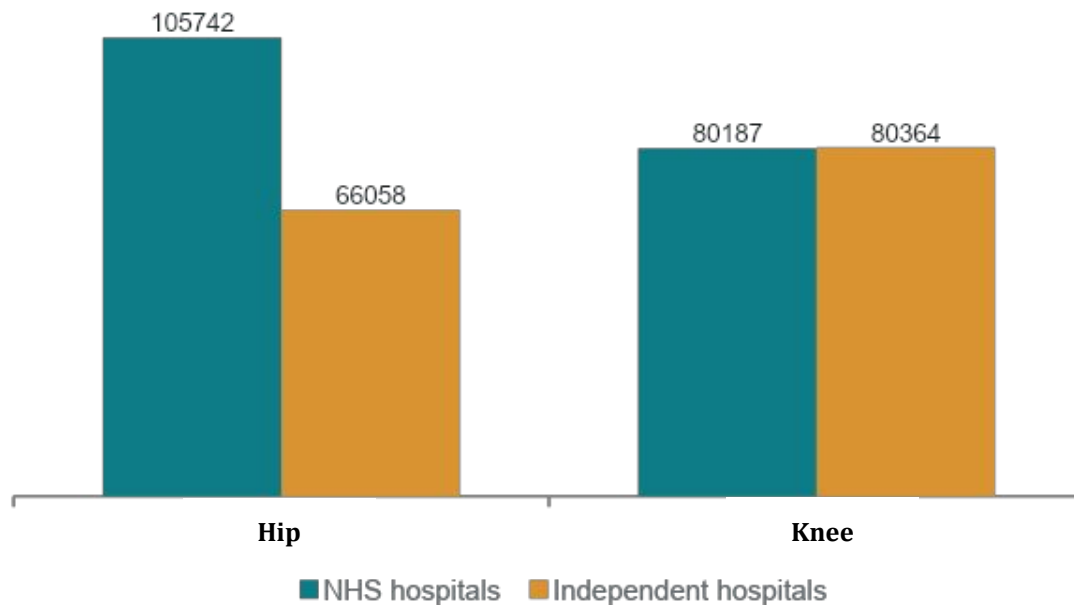


The gap: episodic snapshots, not continuous, objective recovery data.

A large, fast-shifting recovery pathway

UK hip & knee procedure submissions, 2025

National Joint Registry — StatsOnline



332,351

recorded UK hip & knee submissions in 2025 (NHS + independent registry activity)



2.75 / 2.91 days

median length of stay for hip / knee replacement by 2022 — recovery has shifted home



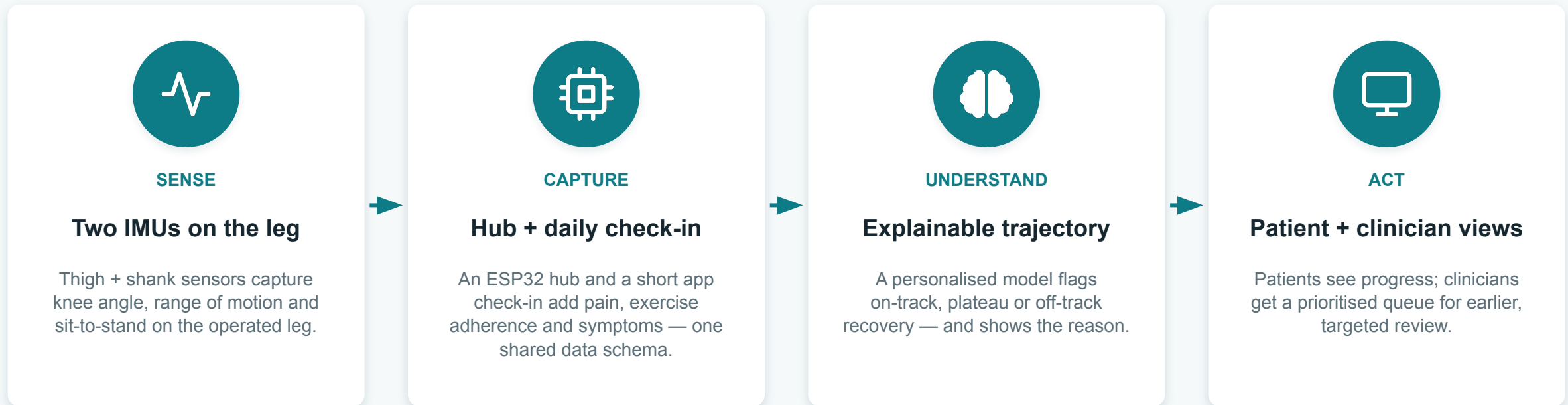
10 million

projected global shortage of health workers by 2030 — systems must do more with less

NICE recommends self-directed rehab after discharge, **with supervised review only when goals aren't met** — exactly the decision RecoverAI Joint supports.

Sources: NJR StatsOnline 2025 · BMC Medicine 2025 · WHO. Aggregate registry data — used to size the problem, not to train the model.

RecoverAI Joint: from movement to clinical insight



4 THINGS THAT MAKE IT WORK

✓ Objective movement

✓ Daily symptoms

✓ Explainable trajectory

✓ Two interfaces

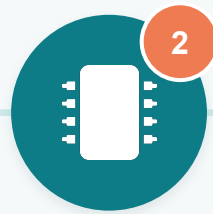
One line: a low-cost wearable and app turn home rehabilitation into an explainable recovery trajectory that tells orthopaedic teams who needs a closer look.

How it works — in five simple steps



Sensors detect movement

You wear two small sensors above and below the knee.



Data goes to the hub

The sensors send readings to a small central hub.



Angle & metrics calculated

The hub works out your knee angle, ROM and reps.



AI checks your trend

It compares today with your expected recovery path.



The app shows feedback

You see your progress; your clinician sees a summary.



If your recovery is behind — or something looks unusual — the system alerts your physiotherapist or doctor.

Where the sensors go



THIGH MODULE

Front of the thigh, about 10–15 cm above the centre of the kneecap.

SHANK MODULE

Front of the shin, about 10–15 cm below the centre of the kneecap.

Same operated leg only.

Nothing is placed on the kneecap, on the surgical wound, or inside the joint. The pods simply move with the leg — no needles, no electrodes.

WHY TWO SENSORS?

Knee angle is the difference between the two segments. A single sensor can't tell knee bending from the whole leg moving.

$$\theta_{knee} = \theta_{shank} - \theta_{thigh} - \theta_{cal}$$

How to wear & install

1. Attach Thigh Module	2. Attach Shank Module	3. Connect Cables	4. Power On	5. Calibrate	6. Start Exercise
					
Place above the knee on the front of thigh.	Place below the knee on the front of shin.	Plug both modules into the central hub.	Press the power button (LED on).	Keep your leg straight and press Calibrate.	Do your exercises. Data is recorded automatically.

✓ No needles

✓ No surgery

✓ Reusable

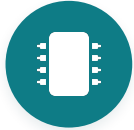
✓ Comfortable

✓ Easy to use



Designed for patients at home — elastic straps or a compression-sleeve pocket keep alignment repeatable. Press start once and data records automatically; the phone becomes the main display.

Low-cost, reusable — 3 simple parts



Sensor modules ×2

Adafruit LSM6DSOX 6-DoF breakout — the two motion sensors.

≈ 25 × 20 mm each



Central hub ×1

ESP32 + Li-Po battery + microSD (plus dev OLED & button).

≈ 55 × 35 mm



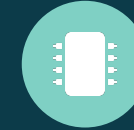
Cable ×2

Thin flexible JST 6-pin, routed along the outside of the leg.

one thin flexible lead



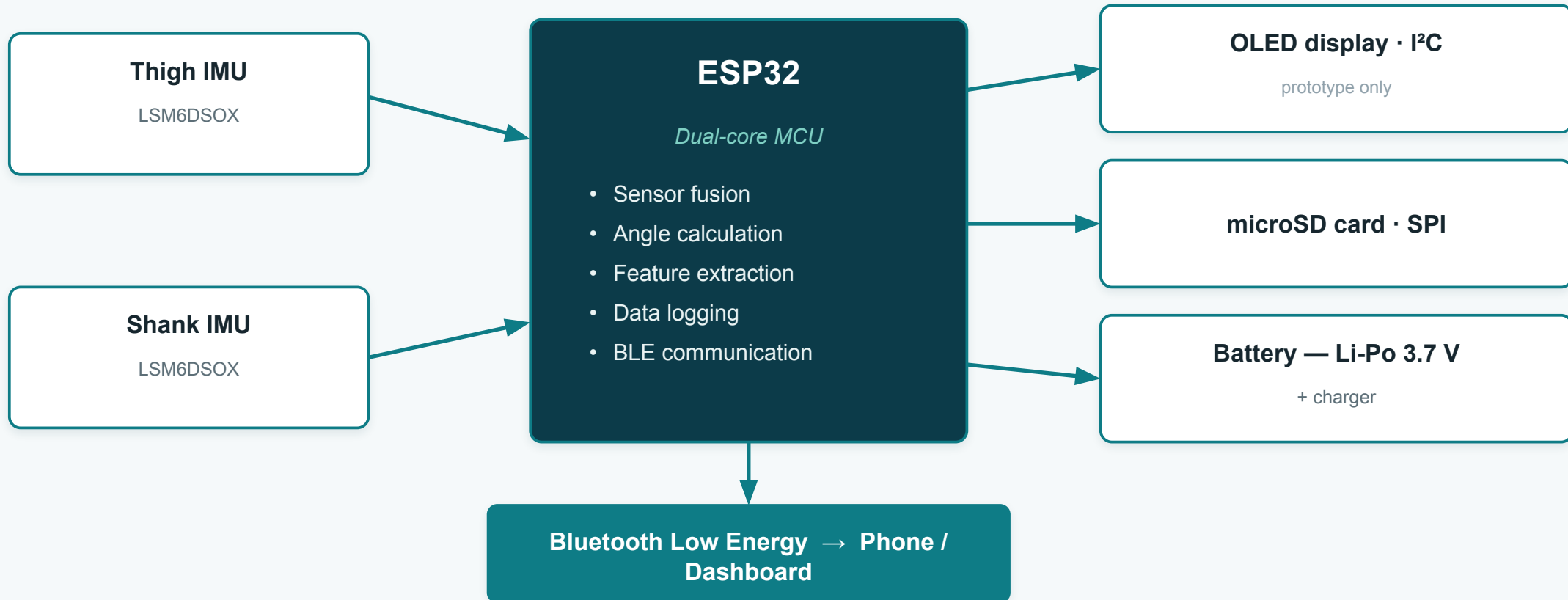
A development-board prototype is enough to win the competition — a custom PCB is the next step (see the KiCad slide).



WHY LSM6DSOX?

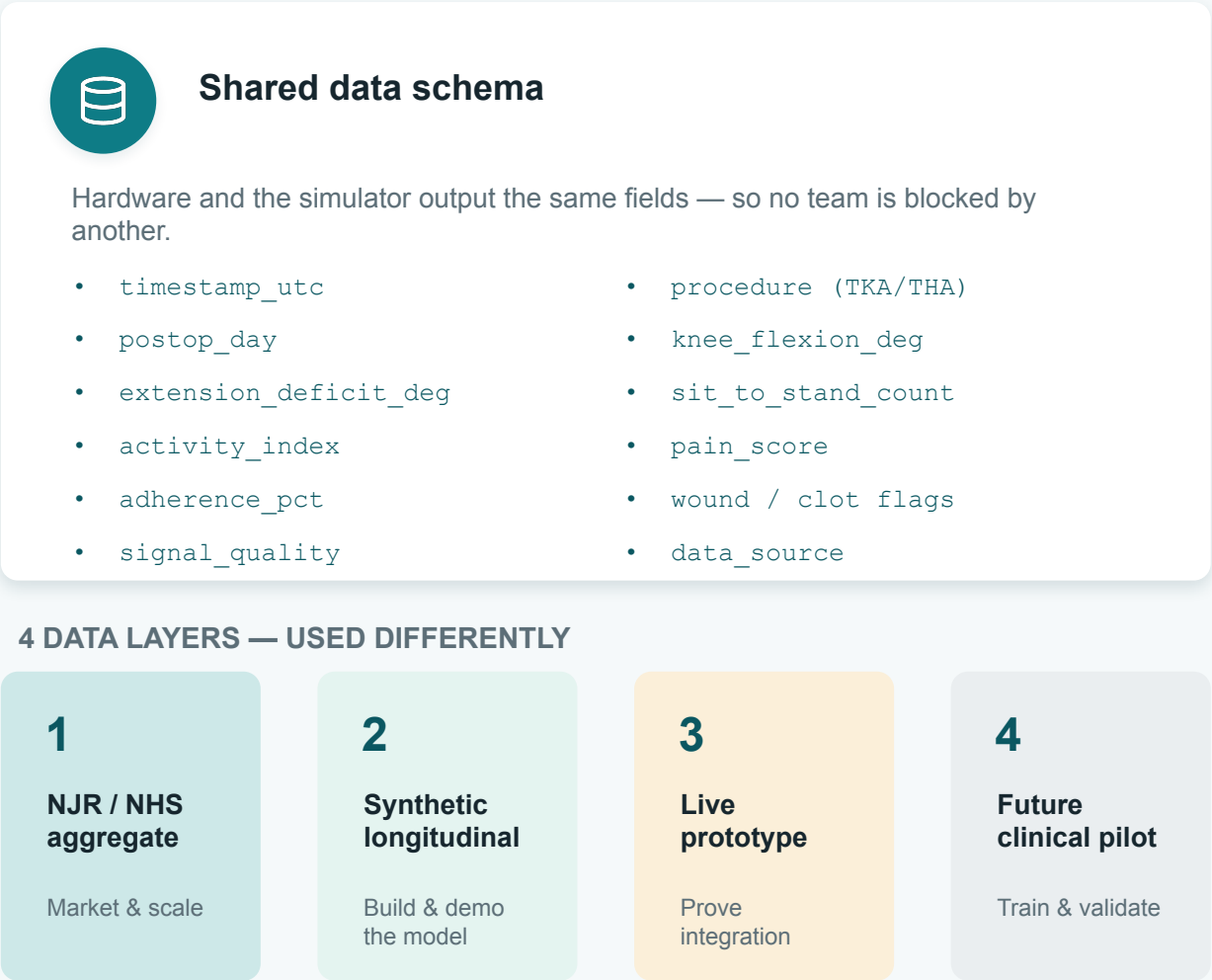
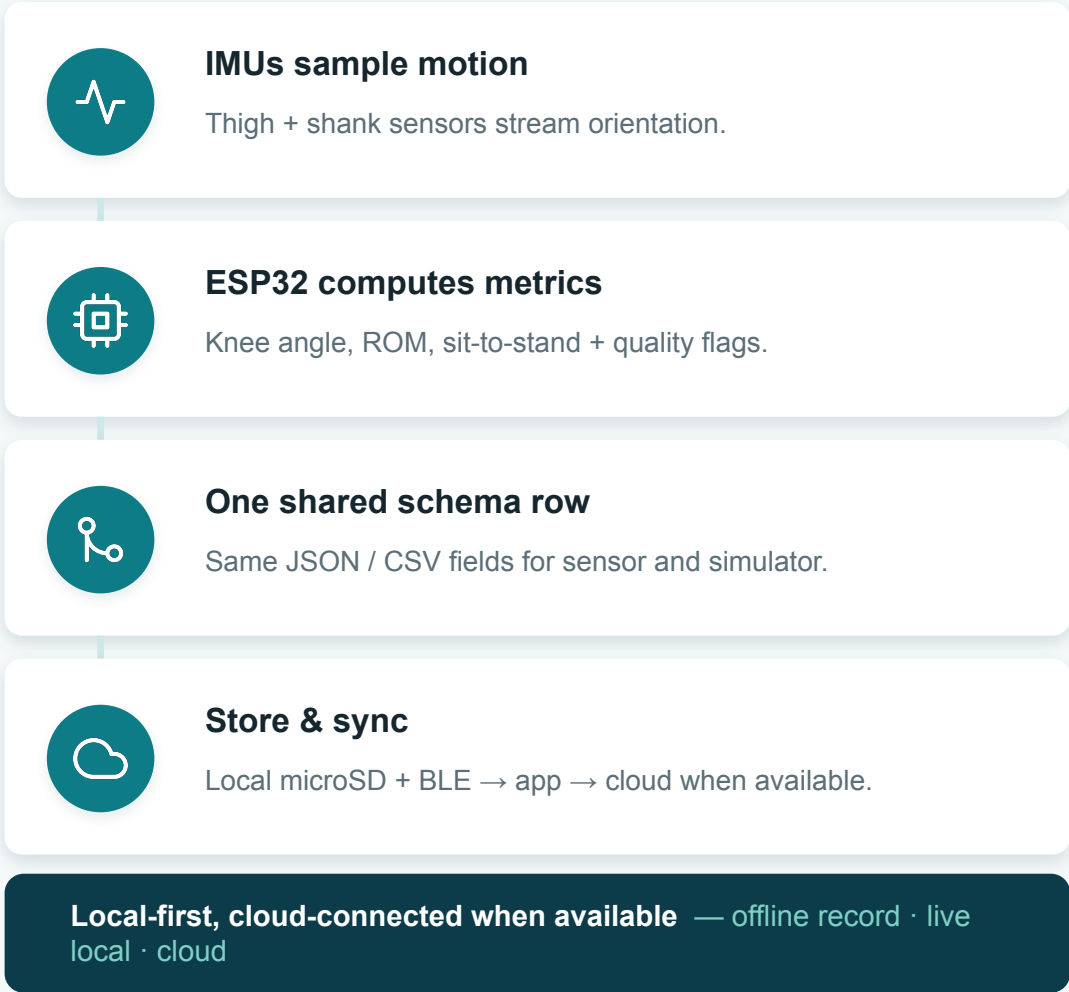
- 6-axis IMU — accelerometer + gyroscope in one chip
- High performance, low power (~0.55 mA)
- Built-in ADC, digital filters & FIFO
- I²C / SPI digital output — no noisy analog front-end
- ±16 g accel · ±2000 dps gyro
- Small & reliable — widely used in wearable medical & sports devices

Electronic block diagram

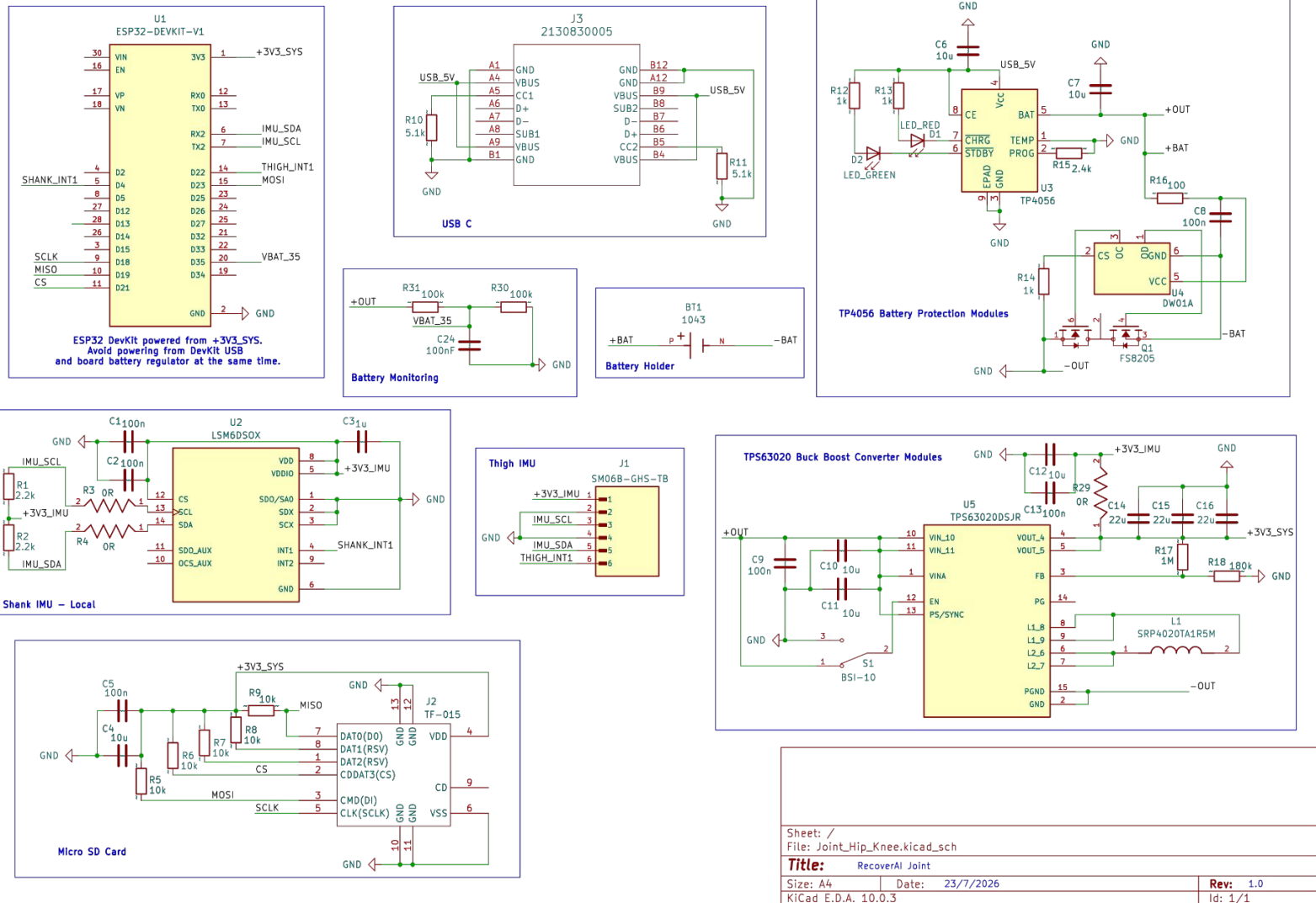


OLED + joystick/button are prototype-only, useful for bench testing. In the final wearable the phone replaces them — one button + one status LED + the app.

From raw motion to a shared data contract



KiCad schematic

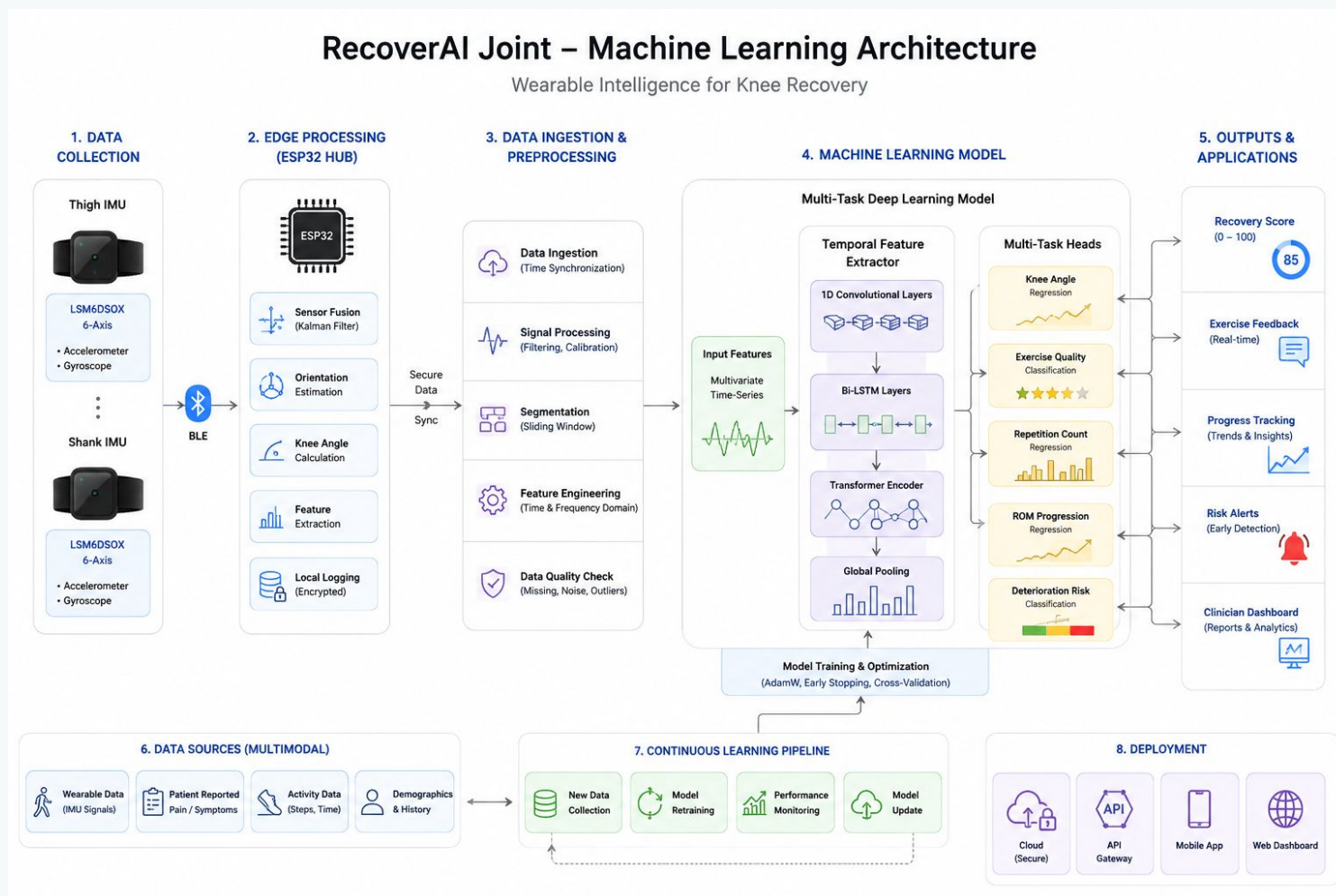


WHAT TO INCLUDE

- ESP32 core + programming / USB
- Dual LSM6DSOX on I²C (0x6A / 0x6B) + INT lines
- 3.7 V LiPo → 3.3 V regulator power tree
- JST 6-pin shank connector: +3V3, GND, SDA, SCL, INT, spare
- 100 nF decoupling caps at each IMU
- Status LED + calibration / start button
- microSD (SPI) for offline logging

Owner · Tommy Tse (Hardware) — target 24 Jul

Recovery-trajectory model



WHAT TO SHOW

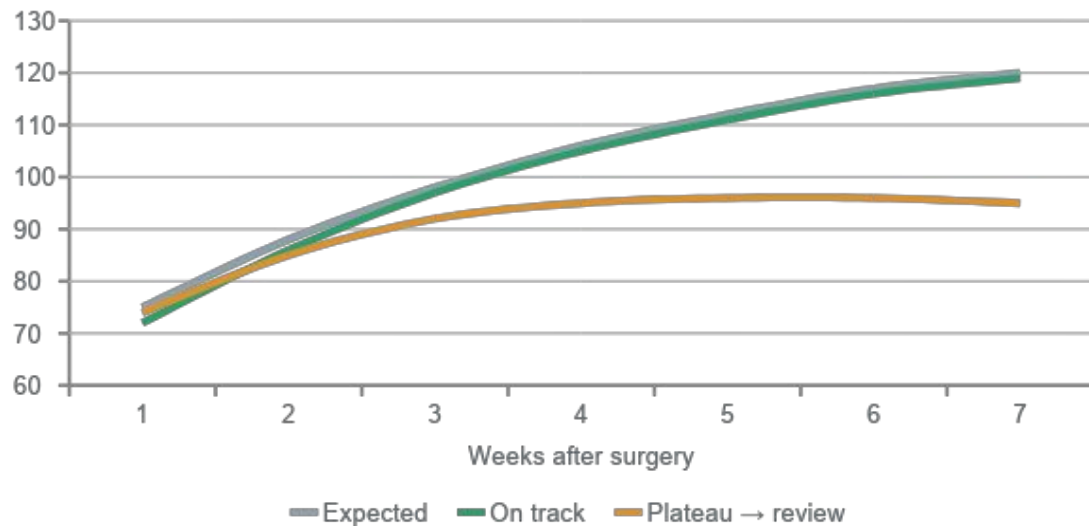
- Align each patient by procedure & post-op day
- Start from expected synthetic curves + patient baseline
- Deviation / plateau / reversal / persistence per metric
- Gate on data quality before combining signals
- Output: on-track / plateau / off-track + explanation
- Evaluate: sensitivity, specificity, false-alert rate, lead time

Explainable before complex — a transparent 'glass-box' model that works end-to-end beats an unvalidated deep net.

Owner · Swapnil Kumar (Software & ML) — target 24 Jul

Explainable recovery trajectory & alerts

Knee flexion vs the expected recovery path



REASON PANEL — why an alert appeared

Flexion plateaued for 4 readings · sit-to-stand slowed 22%
· pain 3 → 6

→ **Amber: non-urgent clinical review recommended**



GREEN — On track

Within the expected band, data quality OK → continue the plan.



AMBER — Review recommended

≥ 2 measures plateau/reverse, or pain/adherence worsens → non-urgent review.



RED — Safety route

Urgent NHS red-flag symptoms → standard urgent-care guidance. Rule-based, not a diagnosis.



GREY — No reliable data

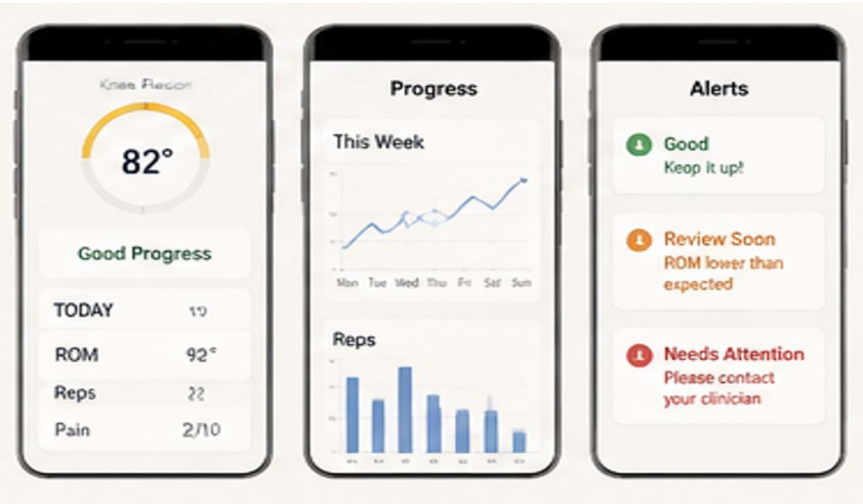
Sensor missing / poor placement → check sensor or guided measurement.

What patients and clinicians see



Patient interface

- Today — prescribed exercise, brief symptom check, sensor status
- Progress — flexion, extension, sit-to-stand & activity trends
- Guidance — safe educational content + point of contact
- Accessibility — large text, simple language, web fallback



Clinician interface

- Priority queue — green / amber / red-safety / no-data groups
- Patient timeline — actual versus expected trajectory
- Reason panel — the exact variables behind each flag
- Data-quality panel — wear time, missing data & battery

21	1	LOW	23002	337	0.8949	0	0	6.3	100
22	1	LOW	23012	347	0.8716	0	0	6.3	100
23	1	LOW	23042	377	0.8705	0	0	6.3	100
24	1	LOW	23052	387	0.8141	0	0	6.3	100
25	1	LOW	23062	397	0.771	0	0	6.3	100
26	1	LOW	23092	427	0.7806	0	0	6.3	100
27	1	LOW	23102	437	0.638	0	0	6.3	100
28	1	LOW	23112	447	0.6308	0	0	6.3	100
29	1	LOW	23142	477	0.6396	0	0	6.3	100
30	1	LOW	23152	487	0.6517	0	0	6.3	100
31	1	LOW	23162	497	0.6492	0	0	6.3	100
32	1	LOW	23192	527	0.642	0	0	6.3	100
33	1	LOW	23202	537	0.5738	0	0	6.3	100
34	1	LOW	23212	547	0.6391	0	0	6.3	100
35	1	LOW	23242	577	0.621	0	0	6.3	100
36	1	LOW	23252	587	0.5906	0	0	6.3	100
37	1	LOW	23262	597	0.6084	0	0	6.3	100
38	1	LOW	23292	627	0.6496	0	0	6.3	100
39	1	LOW	23302	637	0.6112	0	0	6.3	100
40	1	LOW	23312	647	0.567	0	0	6.3	100

Try it yourself



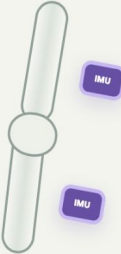
RecoverAI JOINT

Launch application

From movement to a meaningful signal.

Two small motion modules capture the exercise. The central hub calculates useful metrics. The app makes the result understandable.

01

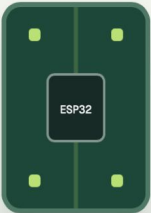


Wear

Place one sensor above the knee and one below it on the front of the leg.

About 10–15 cm from the kneecap

02

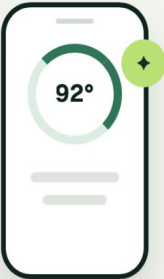


Move

Complete prescribed exercises at home while the hub records motion automatically.

ESP32 hub · battery · local logging

03



Understand

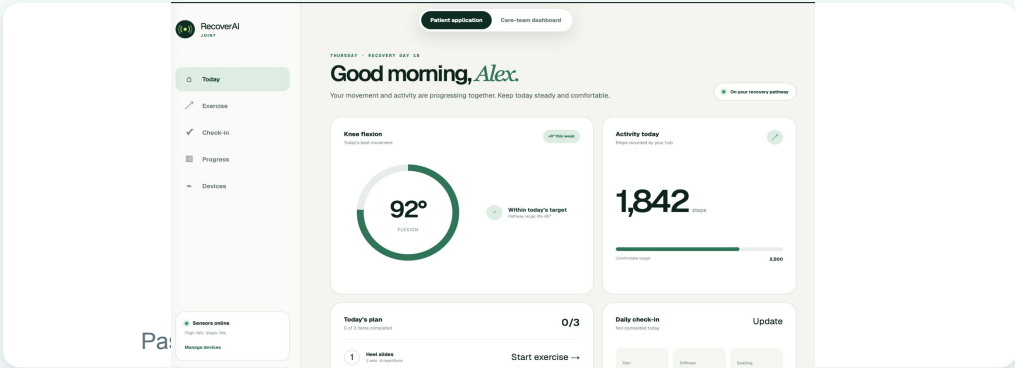
Sensor fusion translates motion into angle, repetition and exercise-quality trends.

Clear feedback for patient and care team



WEB / APP LINK

<https://recoverai-joint-demo.swapnilsrivastava199.chatgpt.site/>



Screen-recording fallback ready (per playbook) in case live hosting or connectivity fails.

Owner · Web/App team — live link by 25 Jul

Why this isn't 'just another recovery app'



Joint-specific movement

Objective knee ROM, extension deficit & sit-to-stand — not just steps or questionnaires.



Explainable prioritisation

Every alert shows the exact measurements and the persistence behind it.



Data-quality awareness

A distinct 'grey' state means bad data never becomes a false clinical alert.



Device-agnostic & low-cost

Reusable sensor pod, phone-based, not tied to one implant vendor.



AI / ML leveraged: personalised time-series trajectory modelling over multi-signal recovery data, with quality gating and human-readable explanations — advancing UN SDG 3 & SDG 9.

How we compare

 Digital orthopaedic recovery is established, but continuous physiological monitoring remains an unmet need.

Platform	Mobility/ROM	Wound signal	Implant agnostic	Primary digital offering
myrecovery (NHS)	Education + PROMs	✗	✓	Education, exercises, pain scores, surveys & dashboards
Zimmer Biomet	✓	✗	✗	ZBEdge / mymobility / Persona IQ
Smith+Nephew / Huma	✓	✗	✓	Remote preparation & monitoring around hip/knee surgery
Specialist gait platforms	✓	✗	✓	Detailed gait analytics & recovery benchmarking
Stryker	✓	Photo upload only	✗	MotionSense / OrthologIQ
Exactech	✓	✗	✗	Muvr
J&J DePuy	App / check-ins	✗	✗	VELYS Insights / Patient Path
RecoverAI Joint	Prototype	Roadmap	✓	External IMUs + symptom reporting

SWOT analysis

STRENGTHS

- Interdisciplinary team: hardware, AI, software, clinical & business
- Objective, joint-specific sensing
- Explainable, safety-first design
- Low-cost, reusable hardware
- Clear NHS workflow fit

WEAKNESSES

- Prototype not yet clinically validated
- Two-IMU placement variability
- No patient-level training data yet
- Depends on patient adherence & digital access

OPPORTUNITIES

- Shorter stays push recovery home
- NHS drive to cut low-value follow-ups
- Expansion to hip & wider MSK surgery
- Integrate with existing wearables & NHS systems

THREATS

- Established competitors (myrecovery, mymobility, Huma)
- MHRA & data-governance burden
- Alert fatigue if false alarms are high
- Reimbursement & procurement uncertainty

Who buys it — and why it matters



Business model

BUYER

NHS trust orthopaedic service (private hospital as earlier pilot)

USER

Orthopaedic physiotherapist / nurse / virtual follow-up team

OFFER

Per-patient pathway software licence + optional reusable sensor pod

PRICING

£50–£100 per monitored pathway — a hypothesis to test with buyers

EXPANSION

Hip-specific sensing, revision/partial knee, wider MSK recovery



Impact

- SDG 3 & 9 — objective, equitable home recovery; upgrades the rehab pathway's technology
- Pilot metrics — adherence, time-to-review, unscheduled contacts, PROMs, false-alert burden, clinician time
- Reduces clinician burden through a prioritised queue — not more admin



Illustrative market scenario

~ £16.6m – £33.2m annual pool if applied to all 332,351 recorded 2025 submissions.

Top-down scenario only — before adoption, discounts, hardware, support & eligibility. Not a revenue claim.

RecoverAI Joint — built for this

**Giada Baroni**

Product & Vision Lead · proposed
Team Leader

**Tommy Tse**

Hardware, Biosignals & Systems
Lead

**Swapnil Kumar**

Software & ML Build Lead

**Ananya Swain**

Clinical, Regulatory &
Data-Governance Lead

**Bhavyansh S. Girdhar**

Commercial & Go-to-Market Lead

**Sareh Moradi**

Advanced-Sensing Research Lead

**Interdisciplinary by design**

≥ 2 faculties — engineering, computing, clinical & business — mapped to three
parallel workstreams: hardware · AI & clinical · app & business.

**To scale, we'd add**

- A clinical-validation partner (orthopaedic physio / PI)
- A regulatory & QMS advisor (MHRA, DCB0129/0160)
- A UX / accessibility specialist for older users

Responsible by design



What we do NOT claim

RecoverAI Joint is a clinical decision-support prototype — not a diagnostic medical device.

- It does not diagnose infection, DVT, pulmonary embolism or implant failure
- It does not predict or guarantee reduced readmission
- It never downgrades a severe symptom because movement data looks normal



Red-flag safety route

Urgent NHS red-flag symptoms bypass the recovery score and show standard urgent-care guidance — contact the care team, NHS 111 or 999. The score never overrules that advice.



Four-layer data plan

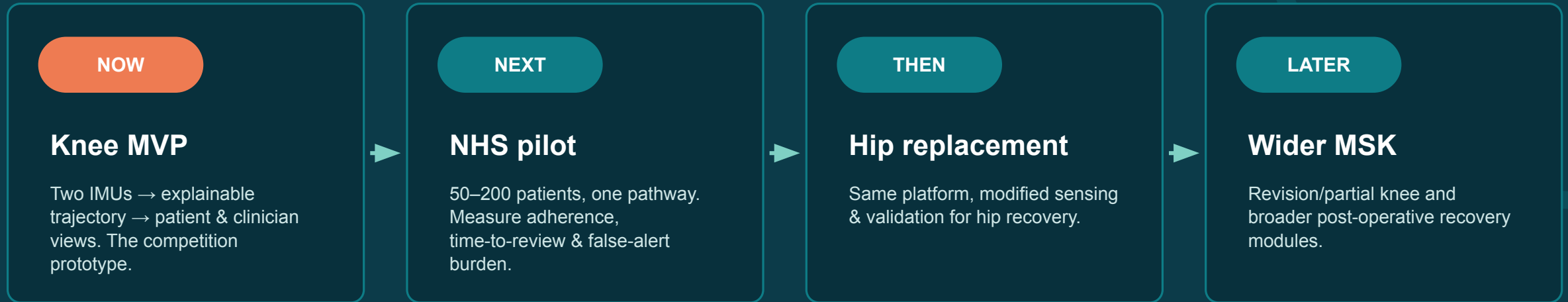
- NJR / NHS aggregate — market sizing, not training data
- Synthetic longitudinal — build & demonstrate the model
- Live prototype — prove hardware-to-dashboard integration
- Future consented pilot — train & validate with clinical labels



UK pathway checkpoints

- Intended purpose first — determines medical-device status
- UK GDPR special-category health data + DPIA for higher-risk processing
- NHS clinical safety — DCB0129 (manufacture) & DCB0160 (deployment)
- MHRA route for any future diagnostic / treatment claim

Knee-first today. Hip & beyond tomorrow.



RecoverAI Joint — so every joint-replacement patient recovers safely at home, and no warning sign is missed.

Thank you · Team OneHealth